

Hybrid LSMC-PDE for a Generalized Gatheral Double Mean-Reverting Model

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Abstract

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We study Bermudan-style optimal stopping problems under a generalized version of Gatheral’s double mean-reverting stochastic volatility model. Our first contribution is a well-posedness analysis for the two-factor variance system: we identify parameter regimes (in particular $\delta_1, \delta_2 \geq 1/2$ under natural nonnegativity assumptions) ensuring existence of a nonnegative strong solution, which is essential for a stochastic-volatility interpretation. Our second contribution connects this model to the Mixed Least-Squares Monte Carlo-partial differential equation (LSMC-PDE) given in Farahany et al. [10] for Bermudan options. We show that the generalized double mean-reverting model satisfies the structural hypotheses needed to (i) represent conditional continuation values via lower-dimensional (random-coefficient) parabolic PDEs and (ii) obtain almost-sure convergence of the regression coefficients under standard truncation/separability conditions.

Keywords: Bermudan options, Hybrid LSMC-PDE, Gatheral Double Mean-Reverting Model

1 Introduction

Options are derivative contracts that give the holder the right, but not the obligation, to buy or sell an underlying asset at a pre-agreed strike price. Under standard no-arbitrage assumptions, contingent claims can be valued under an equivalent risk-neutral measure as discounted expectations of future payoffs. In continuous time, this viewpoint is foundational in the classical diffusion-based option pricing literature and leads to the celebrated Black–Scholes and Merton frameworks [1, 20]. For Markov diffusion models, the probabilistic valuation principle is equivalent to solving a parabolic partial differential equation (PDE) via the Feynman–Kac representation [16].

For European-style contracts under Markov dynamics, the price solves a linear parabolic PDE, while early-exercise features lead to a free-boundary/variational inequality formulation. Finite-difference and tree-based discretizations are effective in low dimensions; the Cox–Ross–Rubinstein binomial model provides a canonical discrete-time framework that yields an efficient numerical procedure and recovers Black–Scholes in the limit [6].

Monte Carlo estimators scale favorably with state dimension and are natural for path-dependent payoffs, since they approximate the risk-neutral expectation directly [13].

Early-exercise contracts (American and Bermudan options) require additional structure. American options may be exercised at any time up to maturity, whereas Bermudan options restrict exercise to a finite set of dates; Bermudan contracts therefore form a natural discretization of the American problem and arise directly in many structured products. Mathematically, American/Bermudan valuation can be formulated as an optimal stopping problem under the risk-neutral measure, where the key computational object is the *continuation value*, i.e. a conditional expectation of future optimal payoffs. In PDE form, early exercise leads to a free boundary or variational inequality; foundational treatments and justifications of classical algorithms for American puts include [21, 15]. In simulation form, the main difficulty is approximating conditional expectations from sampled paths; regression-based approximate dynamic programming methods such as the Longstaff–Schwartz least-squares Monte Carlo (LSMC) approach [19] and the related Tsitsiklis–Van Roy method [23] have become standard, with convergence analysis developed in [4].

In recent years, *hybrid* Monte Carlo–PDE methods have re-emerged for stochastic volatility models with *one-way coupling*, where volatility factors can be simulated independently of the asset. Conditional on a volatility path over a time step, one solves a lower-dimensional PDE in the asset variable to produce a smooth “pre-surface”

in the asset direction, and then uses regression across volatility states to complete the continuation-value surface. This perspective appears in mixed PDE/Monte Carlo developments for stochastic volatility models [18, 17, 8] and is formalized in the mixed LSMC–PDE framework of Farahany et al. [10], who prove almost-sure convergence of regression coefficients for a class of pure stochastic-volatility models under truncation and separability assumptions.

The present work studies Bermudan-style optimal stopping problems under a generalized version of Gatheral’s double mean-reverting stochastic volatility model. The classical double mean-reverting specification features an instantaneous variance v_t that mean-reverts to a stochastic level v'_t , which itself mean-reverts to a constant θ , with square-root (CIR-type) diffusion coefficients. Here we replace the square-root coefficients by CEV-type powers with exponents (δ_1, δ_2) , obtaining a “double CEV” variance system. This generalization raises well-posedness questions near the boundary at 0 because the diffusion coefficients are non-Lipschitz when $\delta_i < 1$; the regime $\delta_i \geq 1/2$ is natural from the standpoint of pathwise uniqueness for CEV-type diffusions [24] and aligns with Gatheral’s “double CEV” modeling perspective [11]. Recent work also motivates reflected variants to mitigate boundary sticking effects in double mean-reverting volatility factors [22].

Our main goal is to place the generalized double mean-reverting (gDMR) model within the hybrid LSMC–PDE convergence theory of Farahany et al. [10]. Concretely, we (i) establish strong well-posedness and nonnegativity of the variance factors in the principled exponent regime and (ii) verify that the gDMR dynamics satisfy the structural hypotheses needed to derive conditional PDE representations and almost-sure convergence of the hybrid regression coefficients.

2 Literature review

Bermudan and American option valuation can be formulated as optimal stopping problems under a risk-neutral measure, where the key numerical task is approximating the conditional continuation values across exercise dates. Regression Monte Carlo methods address this by simulating state trajectories and approximating conditional expectations through least-squares projections onto a prescribed function class. The two canonical formulations are the least-squares Monte Carlo (LSMC) algorithm of Longstaff–Schwartz [19] and the regression-based approximate dynamic programming approach of Tsitsiklis–Van Roy [23]. A rigorous analysis of regression-based estimators in a setting closely aligned with LSMC was provided by Clément, Lamberton, and Protter [4], which clarifies how approximation error and sampling noise propagate through the backward dynamic programming recursion.

While regression Monte Carlo is flexible and scalable, its performance can deteriorate in stochastic-volatility settings, where the continuation value may vary sharply with the volatility state and exercise boundaries may become difficult to learn globally in the enlarged state space. Mixed Monte Carlo–PDE methods mitigate this challenge by conditioning on simulated “background” factors and computing the remaining conditional expectations with deterministic solvers, thereby reducing both dimensionality and variance. In multi-factor FX applications, Cozma and Reisinger [8] developed a mixed MC/PDE variance-reduction framework that exemplifies this conditioning philosophy in a high-dimensional setting. For Bermudan options, Farahany, Jackson, and Jaimungal [10] introduced a hybrid LSMC–PDE paradigm specialized to *one-way coupled* stochastic volatility models, where volatility evolves autonomously and the asset is conditionally lognormal given the variance path. Under explicit truncation and separability conditions, they establish almost-sure convergence of the regression coefficients; additional algorithmic developments and motivation for mixed MC–PDE ideas in higher-dimensional optimal stopping are given in Farahany’s thesis [9].

On the modeling side, accurately capturing equity index option surfaces across strikes and maturities often requires moving beyond single-factor stochastic volatility models such as Heston’s specification [14], in which variance follows a square-root (Cox–Ingersoll–Ross) diffusion [7]. Empirical evidence that multi-factor volatility improves the fit to the equity index “smirk” and its term structure motivates models with multiple variance factors; see, for example, Christoffersen, Heston, and Jacobs [3]. Within the “variance curve” perspective aimed at joint SPX/VIX option consistency, Gatheral proposed a *double mean-reverting* structure in which short variance mean-reverts to a stochastic level that itself mean-reverts more slowly, and advocated a flexible “Double CEV” class where volatility-of-volatility coefficients follow power-law forms with exponents in $[1/2, 1]$. This motivates generalized double mean-reverting models with CEV-type diffusion terms, extending the classical double square-root case and connecting variance-factor dynamics to the broader mean-reverting CEV/CKLS family used in interest-rate modeling [2] and to the classical CEV literature [5].

These generalized variance specifications raise important well-posedness issues near the boundary at 0. For diffusion coefficients that are only Hölder continuous, strong existence and pathwise uniqueness typically rely on Yamada–Watanabe-type criteria [24], which helps explain the frequent restriction $\delta \geq 1/2$ in power-law diffusion models. Recent work has also emphasized the practical and statistical consequences of variance factors spending long times near 0 in Gatheral-type double mean-reverting models, and proposes Skorokhod-reflected variants to

prevent boundary sticking while preserving the mean-reverting structure [22]. For broader background on volatility surface modeling and multi-factor stochastic volatility constructions used by practitioners, see Gatheral [12].

3 Model and Bermudan stopping problem

3.1 Stochastic basis and generalized gDMR dynamics

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{Q})$ be a filtered probability space satisfying the usual conditions and carrying a 3-dimensional $(\mathcal{F}_t)$ -Brownian motion $W = (W^{(1)}, W^{(2)}, W^{(3)})$ with constant correlation matrix $\rho = (\rho_{ij})_{1 \leq i, j \leq 3}$, assumed symmetric positive semidefinite with $\rho_{ii} = 1$. Fix parameters $r \geq 0$ and $\kappa_1, \kappa_2, \theta, \xi_1, \xi_2 > 0$ and initial conditions $(S_0, v_0, v'_0) \in (0, \infty) \times [0, \infty) \times [0, \infty)$.

We consider the generalized Gatheral double mean-reverting (gDMR) system under $\mathbb{Q}$:

$$\begin{aligned} dS_t &= r S_t dt + S_t \sqrt{v_t} dW_t^{(1)}, \\ dv_t &= \kappa_1 (v'_t - v_t) dt + \xi_1 v_t^{\delta_1} dW_t^{(2)}, \\ dv'_t &= \kappa_2 (\theta - v'_t) dt + \xi_2 (v'_t)^{\delta_2} dW_t^{(3)}, \end{aligned} \tag{1}$$

with elasticity parameters $\delta_1, \delta_2 > 0$.

Remark 3.1 (One-way coupling). The variance subsystem (v, v') is autonomous (it does not depend on S). This one-way coupling is the structural feature exploited by mixed MC–PDE methods: volatility paths can be sampled independently of the asset, after which asset expectations are computed conditionally [10].

4 Well-posedness

4.1 Exponent regime and strong well-posedness

Assumption 4.1 (CEV exponents). We impose $\delta_1, \delta_2 \in [1/2, 1]$.

Theorem 4.2 (Strong existence/uniqueness and positivity). *Assume Assumption 4.1. Then for any $T > 0$ the system (1) admits a unique strong solution $(S_t, v_t, v'_t)_{t \in [0, T]}$ with continuous paths such that*

$$S_t > 0, \quad v_t \geq 0, \quad v'_t \geq 0, \quad \forall t \in [0, T] \quad \mathbb{Q}\text{-a.s.}$$

Proof. **Well-posedness and nonnegativity of v' .** Define globally measurable coefficients on $\mathbb{R}$ by

$$b_2(x) := \kappa_2(\theta - x), \quad \sigma_2(x) := \xi_2(x^+)^{\delta_2}, \quad x^+ := \max(x, 0).$$

Then b_2 is globally Lipschitz with linear growth, and σ_2 is Hölder of order δ_2 . For $\delta_2 \geq 1/2$, the Yamada–Watanabe modulus condition holds, implying pathwise uniqueness for the one-dimensional SDE

$$dv'_t = b_2(v'_t)dt + \sigma_2(v'_t)dW_t^{(3)}, \quad v'_0 \geq 0;$$

see Yamada and Watanabe [24] and Karatzas and Shreve [16, Ch. 5]. Since the coefficients are continuous with linear growth, weak existence holds, and hence a unique strong solution exists.

Well-posedness and nonnegativity of v given v' . Given the adapted continuous process v' , define

$$b_1(t, x) := \kappa_1(v'_t - x), \quad \sigma_1(x) := \xi_1(x^+)^{\delta_1}.$$

For each (t, ω) the map $x \mapsto b_1(t, x)$ is globally Lipschitz with constant κ_1 , and σ_1 is Hölder of order $\delta_1 \geq 1/2$. The same Yamada–Watanabe argument (now with time-dependent but Lipschitz drift) yields a unique strong solution v to

$$dv_t = b_1(t, v_t)dt + \sigma_1(v_t)dW_t^{(2)}.$$

Explicit solution for S and strict positivity. With v nonnegative and progressively measurable, the asset SDE is linear in S and admits the unique strong solution

$$S_t = S_0 \exp\left(\int_0^t \left(r - \frac{1}{2}v_s\right)ds + \int_0^t \sqrt{v_s} dW_s^{(1)}\right),$$

which is strictly positive for all $t \in [0, T]$ whenever $S_0 > 0$. □

Remark 4.3 (Why $\delta_i \geq 1/2$ is natural). The lower bound $\delta_i \geq 1/2$ is a classical threshold for pathwise uniqueness in CEV-type diffusions: for

$$dX_t = b(X_t)dt + \sigma(X_t^+)^\delta dW_t,$$

the Yamada–Watanabe modulus condition holds when $\delta \geq 1/2$, while for $\delta < 1/2$ pathwise uniqueness can fail [24]. Since the hybrid MC–PDE convergence results of Farahany et al. [10] are formulated for models admitting strong unique solutions, Assumption 4.1 is a minimal structural restriction that keeps the gDMR model within that framework.

4.2 Bermudan option valuation as a discrete-time stopping problem

Let $\pi = \{0 = t_0 < \dots < t_M = T\}$ be the exercise grid and let $(h_n)_{n=0}^M$ be measurable payoff functions with $h_n : (0, \infty) \rightarrow \mathbb{R}$. For simplicity, we assume that the payoff at time t_n depends only on S_{t_n} . The Bermudan value process $(V_n)_{n=0}^M$ satisfies the Snell recursion

$$V_M = h_M(S_{t_M}), \quad V_n = \max \left\{ h_n(S_{t_n}), e^{-r(t_{n+1}-t_n)} \mathbb{E}^{\mathbb{Q}}[V_{n+1} \mid \mathcal{F}_{t_n}] \right\}, \quad n = M-1, \dots, 0. \quad (2)$$

Since (S_t, v_t, v'_t) is Markov under $\mathbb{Q}$, the continuation value can be written as a deterministic function of the current state. Specifically, define

$$C_n(s, v, v') := e^{-r(t_{n+1}-t_n)} \mathbb{E}^{\mathbb{Q}}[V_{n+1}(S_{t_{n+1}}, v_{t_{n+1}}, v'_{t_{n+1}}) \mid S_{t_n} = s, v_{t_n} = v, v'_{t_n} = v']. \quad (3)$$

Then the one-step continuation value appearing in (2) is given by

$$e^{-r(t_{n+1}-t_n)} \mathbb{E}^{\mathbb{Q}}[V_{n+1} \mid \mathcal{F}_{t_n}] = C_n(S_{t_n}, v_{t_n}, v'_{t_n}) \quad \mathbb{Q}\text{-a.s.}$$

5 Conditional PDE representation

Since the variance factors (v, v') are *one-way coupled* (their dynamics do not depend on S), the entire volatility path over an interval $[t_n, t_{n+1}]$ is measurable with respect to the Brownian drivers that generate (v, v') . After an orthogonalization of the *asset* Brownian motion, the remaining randomness in the log-price evolution on $[t_n, t_{n+1}]$ is carried by a Brownian motion that is independent of those volatility drivers. Conditional expectations over one exercise step therefore reduce to a one-dimensional (time-inhomogeneous) diffusion problem, which is characterized by a backward parabolic PDE with *random (pathwise) coefficients* Farahany et al. [10].

5.1 Orthogonalization on a Bermudan step

Fix an interval $[t_n, t_{n+1}]$ in a Bermudan exercise grid and define

$$\mathcal{G}_n := \sigma(W_s^{(2)}, W_s^{(3)}) : s \in [t_n, t_{n+1}].$$

Since the variance subsystem is one-way coupled and driven by $(W^{(2)}, W^{(3)})$, the path $(v_s, v'_s)_{s \in [t_n, t_{n+1}]}$ is measurable with respect to $\mathcal{F}_{t_n} \vee \mathcal{G}_n$.

Assume $|\rho_{23}| < 1$ and define

$$\Sigma_{23} := \begin{bmatrix} 1 & \rho_{23} \\ \rho_{23} & 1 \end{bmatrix}, \quad \beta := \Sigma_{23}^{-1} \begin{bmatrix} \rho_{12} \\ \rho_{13} \end{bmatrix} = \begin{bmatrix} \beta_2 \\ \beta_3 \end{bmatrix}.$$

Equivalently,

$$\beta_2 = \frac{\rho_{12} - \rho_{13}\rho_{23}}{1 - \rho_{23}^2}, \quad \beta_3 = \frac{\rho_{13} - \rho_{12}\rho_{23}}{1 - \rho_{23}^2}. \quad (4)$$

The conditional variance of $W^{(1)}$ after projection onto $\text{span}\{W^{(2)}, W^{(3)}\}$ is

$$\sigma_\perp^2 := 1 - \begin{bmatrix} \rho_{12} & \rho_{13} \end{bmatrix} \Sigma_{23}^{-1} \begin{bmatrix} \rho_{12} \\ \rho_{13} \end{bmatrix} = \frac{1 - \rho_{12}^2 - \rho_{13}^2 - \rho_{23}^2 + 2\rho_{12}\rho_{13}\rho_{23}}{1 - \rho_{23}^2} \geq 0. \quad (5)$$

Hence there exists a Brownian motion $B^\perp$, independent of $(W^{(2)}, W^{(3)})$, such that on $[t_n, t_{n+1}]$,

$$dW_t^{(1)} = \beta_2 dW_t^{(2)} + \beta_3 dW_t^{(3)} + \sigma_\perp dB_t^\perp. \quad (6)$$

Let $X_t := \log S_t$. Under $\mathbb{Q}$, we have

$$dX_t = \left(r - \frac{1}{2}v_t\right)dt + \sqrt{v_t} dW_t^{(1)}. \quad (7)$$

Substituting (6) into (7) yields, on $[t_n, t_{n+1}]$,

$$dX_t = \left(r - \frac{1}{2}v_t\right)dt + \sqrt{v_t}(\beta_2 dW_t^{(2)} + \beta_3 dW_t^{(3)}) + \sigma_\perp \sqrt{v_t} dB_t^\perp. \quad (8)$$

Define the pathwise correlated shift and the orthogonal log component

$$Z_t^{\text{shift}} := \int_{t_n}^t \sqrt{v_s} (\beta_2 dW_s^{(2)} + \beta_3 dW_s^{(3)}), \quad Y_t := X_t - Z_t^{\text{shift}}. \quad (9)$$

Then $X_t = Y_t + Z_t^{\text{shift}}$ and, on $[t_n, t_{n+1}]$,

$$dY_t = \left(r - \frac{1}{2}v_t\right)dt + \sigma_\perp \sqrt{v_t} dB_t^\perp. \quad (10)$$

In the degenerate case $\sigma_\perp = 0$, the stock Brownian driver lies in the span of the volatility Brownian drivers and, conditional on the volatility path, the propagation of X becomes deterministic.

5.2 Conditional expectation as a random-coefficient PDE

Lemma 5.1. Assume Assumption 4.1 and let $G : \mathbb{R} \times [0, \infty)^2 \rightarrow \mathbb{R}$ be bounded and Borel measurable. Fix $n \in \{0, \dots, M-1\}$ and work on the interval $[t_n, t_{n+1}]$. Let

$$\mathcal{G}_n := \sigma((W_s^{(2)}, W_s^{(3)}) : s \in [t_n, t_{n+1}]),$$

and define

$$Z_n := \int_{t_n}^{t_{n+1}} \sqrt{v_s} (\beta_2 dW_s^{(2)} + \beta_3 dW_s^{(3)}),$$

where β_2, β_3 are given by (4) and $\sigma_\perp$ by (5). For $t \in [t_n, t_{n+1}]$ and $y \in \mathbb{R}$ set

$$u(t, y) := \mathbb{E}^\mathbb{Q} \left[G(Y_{t_{n+1}} + Z_n, v_{t_{n+1}}, v'_{t_{n+1}}) \middle| \mathcal{G}_n, Y_t = y \right],$$

where Y is the orthogonal log-component in (10) with $Y_{t_n} = \log S_{t_n}$.

Then, conditional on $\mathcal{G}_n$, u is the bounded solution of the random-coefficient backward PDE

$$\begin{cases} \partial_t u(t, y) + \left(r - \frac{1}{2}v_t\right) \partial_y u(t, y) + \frac{1}{2} \sigma_\perp^2 v_t \partial_{yy} u(t, y) = 0, & (t, y) \in [t_n, t_{n+1}] \times \mathbb{R}, \\ u(t_{n+1}, y) = G(y + Z_n, v_{t_{n+1}}, v'_{t_{n+1}}), & y \in \mathbb{R}. \end{cases}$$

Proof. By (6)–(10), on $[t_n, t_{n+1}]$ we have

$$dY_t = \left(r - \frac{1}{2}v_t\right)dt + \sigma_\perp \sqrt{v_t} dB_t^\perp,$$

where $B^\perp$ is a Brownian motion independent of $(W^{(2)}, W^{(3)})$. Since the variance subsystem (v, v') is driven by $(W^{(2)}, W^{(3)})$, the entire path $t \mapsto v_t$ on $[t_n, t_{n+1}]$ is $\mathcal{G}_n$ -measurable. Hence, conditional on $\mathcal{G}_n$, Y is a time-inhomogeneous one-dimensional diffusion with *deterministic* coefficients, and the conditional Feynman-Kac formula yields that u solves the stated backward parabolic PDE with terminal condition at t_{n+1} . $\square$

6 Almost-sure convergence of the hybrid regression coefficients

This section adapts the almost-sure convergence theorem of Farahany et al. [10] to the generalized gDMR model (1). Our goal is to (i) restate the regression coefficients in a self-contained manner (in the present two-factor volatility setting), and (ii) verify, in detail, that the structural hypotheses required by the Farahany et al. [10] framework are satisfied.

Throughout, we work under the risk-neutral measure $\mathbb{Q}$ and the time grid $\pi = \{0 = t_0 < t_1 < \dots < t_M = T\}$ used in the Bermudan recursion.

6.1 Regression set-up and truncation conditions

Let $d_B \in \mathbb{N}$ and let

$$\phi = (\phi_1, \dots, \phi_{d_B}) : [0, \infty)^2 \rightarrow \mathbb{R}^{d_B}$$

be a fixed vector of basis functions in the two-factor volatility state (v, v') . Let $\mathcal{D}_v \subset [0, \infty)^2$ be a compact rectangle such that $\text{supp}(\phi) \subset \mathcal{D}_v$, and let $\mathcal{S} \subset (0, \infty)$ be a compact set containing the support of every exercise function h_n (this is the standard spatial truncation used when solving the conditional PDEs).

Idealized continuation functions. Following Farahany et al. [10], define the *idealized* continuation functions by backward induction:

$$c_M(s, v, v') \equiv 0, \quad c_n(s, v, v') := a_n(s) \cdot \phi(v, v'), \quad n = 0, \dots, M-1,$$

where, for $n < M$, the coefficient vector $a_n(s) \in \mathbb{R}^{d_B}$ is obtained by least-squares regression of the next-step value. Define the associated “value functions”

$$f_n(s, v, v') := \max\{h_n(s), c_n(s, v, v')\}, \quad n = 0, \dots, M-1, \quad \text{and} \quad f_M(s, v, v') := h_M(s).$$

For fixed (n, s) , define the conditional least-squares objective

$$H_{n,s}(a) := \mathbb{E}_{\mathbb{Q}} \left[\left(\mathbb{E}_{\mathbb{Q}} [f_{n+1}(S_{t_{n+1}}, v_{t_{n+1}}, v'_{t_{n+1}}) \mid S_{t_n} = s, v_{t_n}, v'_{t_n}] - a \cdot \phi(v_{t_n}, v'_{t_n}) \right)^2 \mid S_{t_n} = s \right]. \quad (11)$$

The (idealized) coefficient vector is any minimizer

$$a_n(s) \in \arg \min_{a \in \mathbb{R}^{d_B}} H_{n,s}(a).$$

Under the truncation conditions below, $H_{n,s}$ is finite and the normal equations yield the explicit representation:

$$a_n(s) = A_n^{-1} b_n(s), \quad A_n := \mathbb{E}_{\mathbb{Q}} [\phi(v_{t_n}, v'_{t_n}) \phi(v_{t_n}, v'_{t_n})^\top], \quad b_n(s) := \mathbb{E}_{\mathbb{Q}} [\phi(v_{t_n}, v'_{t_n}) f_{n+1}(S_{t_{n+1}}, v_{t_{n+1}}, v'_{t_{n+1}}) \mid S_{t_n} = s]. \quad (12)$$

Estimated continuation functions (hybrid LSMC–PDE). Let $\mathbb{Q}'$ denote the inherited sampling measure used to construct i.i.d. copies of the volatility path segments. Under $\mathbb{Q}'$, let $\{(v_{t_k}^j, v'_{t_k}^j)_{k=0}^M\}_{j \geq 1}$ be i.i.d. copies of the volatility skeleton, and let $[v^j, v'^j]_{t_n}^{t_{n+1}}$ denote the corresponding path segment on $[t_n, t_{n+1}]$.

Define the estimated continuation functions recursively by

$$c_M^N \equiv 0, \quad c_n^N(s, v, v') := a_n^N(s) \cdot \phi(v, v'), \quad f_n^N(s, v, v') := \max\{h_n(s), c_n^N(s, v, v')\}, \quad f_M^N := h_M.$$

At time step n , the hybrid algorithm computes the pre-regression conditional expectations (via the conditional PDE) along each sampled volatility segment:

$$\bar{C}_n^{j,N}(s) := \mathbb{E}_{\mathbb{Q}} [f_{n+1}^N(S_{t_{n+1}}, v_{t_{n+1}}^j, v'_{t_{n+1}}^j) \mid S_{t_n} = s, [v^j, v'^j]_{t_n}^{t_{n+1}}].$$

The regression matrix and right-hand side are then

$$A_n^N := \frac{1}{N} \sum_{j=1}^N \phi(v_{t_n}^j, v'_{t_n}^j) \phi(v_{t_n}^j, v'_{t_n}^j)^\top, \quad b_n^N(s) := \frac{1}{N} \sum_{j=1}^N \phi(v_{t_n}^j, v'_{t_n}^j) \bar{C}_n^{j,N}(s),$$

and the (truncated) hybrid regression coefficient is

$$a_n^N(s) := [A_n^N]_R^{-1} b_n^N(s), \quad [A_n^N]_R^{-1} := [A_n^N]^{-1} \mathbf{1}_{\{\| [A_n^N]^{-1} \| \leq R\}}, \quad (13)$$

for a fixed truncation constant $R > 0$ chosen as in the standard truncation scheme. Using $\mathbf{1}_{\{\| [A_n^N]^{-1} \| \leq R\}}$ is ensuring uniform boundedness of the inverse.

- Assumption 6.1** (Truncation conditions). 1. The basis functions $\{\phi_m\}_{m=1}^{d_B}$ are bounded and supported on the compact rectangle $\mathcal{D}_v \subset [0, \infty)^2$.
2. For each n , the exercise function h_n is bounded and has compact support contained in $\mathcal{S} \subset (0, \infty)$.
3. The regression inverse is truncated as in (13) for some $R > 0$, so that $\sup_{n \leq M-1, N \geq 1} \| [A_n^N]_R^{-1} \| < \infty$ holds $\mathbb{Q}'$ -a.s.

Remark Item 2) holds automatically on any finite PDE grid because h_n is evaluated only on the truncated domain $\mathcal{S}$ ($[s_{\min}, s_{\max}]$). For common payoffs without compact support (e.g. put), one may apply a standard cutoff: truncate the payoff beyond a large threshold and extend continuously to 0 outside a slightly larger threshold. Likewise, common bases (polynomials, etc.) may be multiplied by a smooth cutoff so that $\text{supp}(\phi) \subset \mathcal{D}_v$ while keeping $\mathcal{D}_v$.

6.2 Separability and finite-dimensional path statistics

The almost-sure convergence proof of Farahany et al. [10] relies on separability of the asset dynamics and on the fact that the conditional expectations used in the hybrid algorithm depend on each volatility path segment only through a *finite-dimensional* statistic.

Proposition 6.2 (Multiplicative return representation). *Fix $0 \leq t < u \leq T$. The solution of (1) satisfies*

$$S_u = S_t R_{t,u}, \quad R_{t,u} := \exp\left(\int_t^u \left(r - \frac{1}{2}v_s\right) ds + \int_t^u \sqrt{v_s} dW_s^{(1)}\right).$$

In particular, for fixed (t, u) , the factor $R_{t,u}$ does not depend on the value of S_t .

Proof. Apply the explicit stochastic exponential representation for S from Theorem 4.2 on $[t, u]$. $\square$

Assumption 6.3 (Separability conditions). 1. $S_t > 0$ for all $t \in [0, T]$ almost surely.

2. For each exercise step n , if $S_{t_n} = s$ then $S_{t_{n+1}} = s R_n$, where R_n does not depend on s .

3. The exercise functions h_n are continuous with compact support in $(0, \infty)$, and the basis functions ϕ are continuous with compact support in $[0, \infty)^2$.

Comment. Assumption 6.3(1)–(2) is a model property: it holds for (1) by Theorem 4.2 and Proposition 6.2. Assumption 6.3

6.3 Main convergence theorem

Theorem 6.4 (Almost-sure coefficient convergence for the gDMR model). *Assume Assumption 4.1, and Assumptions 6.1 and 6.3. Fix $n \in \{0, \dots, M-1\}$ and $s \in \mathcal{S}$. Let $a_n(s)$ be the idealized coefficient vector defined by (11)–(12), and let $a_n^N(s)$ be the hybrid LSMC–PDE estimator defined by (13) from N i.i.d. volatility samples under the inherited sampling measure $\mathbb{Q}'$. Then*

$$\lim_{N \rightarrow \infty} \|a_n^N(s) - a_n(s)\| = 0 \quad \mathbb{Q}'\text{-almost surely.}$$

Consequently, the estimated continuation functions $c_n^N(s, v, v') := a_n^N(s) \cdot \phi(v, v')$ converge uniformly on the truncation set $\mathcal{D}_v$:

$$\lim_{N \rightarrow \infty} \sup_{(v, v') \in \mathcal{D}_v} |c_n^N(s, v, v') - c_n(s, v, v')| = 0, \quad \mathbb{Q}'\text{-almost surely.}$$

In particular, $f_n^N(s, v, v') = \max\{h_n(s), c_n^N(s, v, v')\}$ converges uniformly to $f_n(s, v, v')$ on $\mathcal{D}_v$.

Proof. We verify the hypotheses of the almost-sure convergence theorem of Farahany et al. [10] in the present two-factor volatility setting.

Pure stochastic volatility / one-way coupling. By construction, the volatility subsystem (v_t, v'_t) is autonomous and does not depend on S . Hence, volatility path segments can be sampled independently of the current asset level, and the inherited sampling space $(\Omega', \mathcal{F}', \mathbb{Q}')$ of Farahany et al. [10] applies.

Well-posedness. Under Assumption 4.1, Theorem 4.2 yields a unique strong solution of (1) with $S_t > 0$ and $v_t, v'_t \geq 0$ on $[0, T]$ almost surely. This provides the basic existence/uniqueness and positivity required by the framework.

Truncation scheme. Assumption 6.1 is exactly the truncation scheme used in [10]: bounded compactly supported payoff/basis functions make the objectives (11) finite, and the truncated inverse $[A_n^N]_R^{-1}$ makes $a_n^N(s)$ well-defined and uniformly bounded in (n, N) .

Separability. Assumption 6.3(1) holds by positivity of S from Theorem 4.2. Assumption 6.3(2) holds by Proposition 6.2 with $R_n = R_{t_n, t_{n+1}}$. Assumption 6.3(3) is imposed by truncation and matches the compact-support setting required in [10].

Apply Farahany et al. Having verified one-way coupling, truncation, separability, and the finite-statistic property, we may invoke Theorem 1 of Farahany et al. [10] (with volatility state $(v_{t_n}, v'_{t_n}) \in [0, \infty)^2$) to conclude that $\|a_n^N(s) - a_n(s)\| \rightarrow 0$ $\mathbb{Q}'$ -almost surely.

Uniform convergence of $c_n^N(s, \cdot, \cdot)$ on $\mathcal{D}_v$ follows immediately from boundedness of ϕ on $\mathcal{D}_v$:

$$\sup_{(v, v') \in \mathcal{D}_v} |(a_n^N(s) - a_n(s)) \cdot \phi(v, v')| \leq \|a_n^N(s) - a_n(s)\| \sup_{(v, v') \in \mathcal{D}_v} \|\phi(v, v')\|.$$

Finally, uniform convergence of f_n^N follows from the 1-Lipschitz property of $x \mapsto \max\{h_n(s), x\}$. $\square$

7 A hybrid LSMC-PDE methodology for the gDMR model

This section summarizes a computational hybrid method for Bermudan pricing under the one-way coupled gDMR model (1). The method follows the mixed LSMC-PDE philosophy of Farahany et al. [10]: (i) simulate the autonomous volatility factors, (ii) compute the *one-step* conditional expectations in the Snell recursion by solving a *one-dimensional* random-coefficient PDE in the log-price variable along each sampled volatility path, producing a smooth *pre-surface* over the asset grid, and (iii) regress these pre-surface values across volatility states to obtain a completed continuation-value surface over the volatility state space.

7.1 Discretization and regression ansatz

Let $\pi = \{0 = t_0 < \dots < t_M = T\}$ be the Bermudan exercise grid and $(h_n)_{n=0}^M$ the (bounded, compactly supported) exercise payoff functions. We fix a spatial grid of asset values

$$\mathcal{S} := \{s_1, \dots, s_{N_S}\} \subset (0, \infty), \quad y_i := \log s_i,$$

which is used by the conditional PDE solver in log-space. We also fix a truncation rectangle $D_v \subset [0, \infty)^2$ in the volatility state space and choose d_B bounded basis functions

$$\phi = (\phi_1, \dots, \phi_{d_B}) \quad \text{supported on } D_v.$$

Following Farahany et al. [10], we approximate the continuation value at time t_n in the separable form

$$c_n(s, v, v') \approx a_n(s) \cdot \phi(v, v'), \quad a_n(s) \in \mathbb{R}^{d_B}. \quad (14)$$

In practice we compute $a_n(s)$ only on the grid $s \in \mathcal{S}$, and use interpolation in s when evaluating the approximation at off-grid asset values.

We truncate the volatility state space by fixing a compact rectangle $D_v = [0, \bar{v}] \times [0, \bar{v}'] \subset [0, \infty)^2$, where $\bar{v}, \bar{v}'$ are chosen as high empirical quantiles of $\{(v_{t_n}^j, (v'_{t_n})^j)\}$ from the volatility simulation.

7.2 Pathwise conditional PDE step and pre-surfaces

Fix a time index $n \in \{0, \dots, M-1\}$ and assume we have already constructed an approximation $\widehat{V}_{n+1}$ of the value function at t_{n+1} on $\mathcal{S} \times D_v$. For each Monte Carlo sample $j \in \{1, \dots, N\}$ we simulate a volatility path segment on $[t_n, t_{n+1}]$,

$$(v_t^j, (v'_t)^j)_{t \in [t_n, t_{n+1}]},$$

using a fine internal time discretization on the interval. Along with the volatility segment we compute the pathwise correlated shift

$$Z_n^j := \int_{t_n}^{t_{n+1}} \sqrt{v_s^j} (\beta_2 dW_s^{(2),j} + \beta_3 dW_s^{(3),j}), \quad (15)$$

where β_2, β_3 are given by (4) and $\sigma_\perp$ by (5). By Lemma 5.1 (Section 5), conditional on the volatility drivers on $[t_n, t_{n+1}]$ the orthogonalized log-price component Y evolves as a one-dimensional diffusion with deterministic coefficients,

$$dY_t = \left(r - \frac{1}{2}v_t^j\right) dt + \sigma_\perp \sqrt{v_t^j} dB_t^\perp, \quad t \in [t_n, t_{n+1}],$$

and the conditional expectation required for the continuation value may be written as the solution of a backward parabolic PDE in (t, y) . Concretely, define the terminal function on log-space by

$$g_{n+1}^j(y) := \widehat{V}_{n+1}(e^y, v_{t_{n+1}}^j, (v'_{t_{n+1}})^j),$$

and let $u_n^j(t, y)$ denote the conditional expectation

$$u_n^j(t, y) := \mathbb{E}^Q \left[g_{n+1}^j(Y_{t_{n+1}} + Z_n^j) \mid Y_t = y, (v^j, (v')^j)_{t_n}^{t_{n+1}} \right].$$

Then u_n^j satisfies, on $[t_n, t_{n+1}) \times \mathbb{R}$,

$$\partial_t u + \left(r - \frac{1}{2}v_t^j\right) \partial_y u + \frac{1}{2} \sigma_\perp^2 v_t^j \partial_{yy} u = 0, \quad u(t_{n+1}, y) = g_{n+1}^j(y + Z_n^j). \quad (16)$$

The *pre-surface* is obtained by evaluating the PDE solution at $t = t_n$ on the spatial grid:

$$\widehat{C}_n^j(s_i) := e^{-r\Delta t_n} u_n^j(t_n, y_i), \quad i = 1, \dots, N_S, \quad (17)$$

where $\Delta t_n = t_{n+1} - t_n$. This provides, for each volatility sample j , a smooth function of $s \in \mathcal{S}$. **Add comment how to solve this using different method**

Because (16) has coefficients depending only on time (through v_t^j), it is translation-invariant in y and admits a closed-form solution in Fourier space (cf. Farahany et al. [10]). Let

$$A_n^j := \int_{t_n}^{t_{n+1}} \left(r - \frac{1}{2} v_s^j \right) ds, \quad B_n^j := \sigma_\perp^2 \int_{t_n}^{t_{n+1}} v_s^j ds.$$

Taking the Fourier transform in y reduces the PDE to an ODE in time, yielding the characteristic exponent

$$\Psi_n^j(\omega) := i\omega Z_n^j + i\omega A_n^j - \frac{1}{2} \omega^2 B_n^j.$$

On a uniform log-grid in y , FST computes the recursion

$$u_n^j(t_n, \cdot) \approx \text{FFT}^{-1} \left(\text{FFT}[g_{n+1}^j] \exp(\Psi_n^j) \right), \quad (18)$$

where the exponential is applied pointwise in Fourier space. In the present one-dimensional setting this produces all values $u_n^j(t_n, y_i)$ simultaneously, after the pathwise scalars (Z_n^j, A_n^j, B_n^j) have been computed from the volatility discretization.

7.3 Regression across volatility states and backward recursion

Having computed the pre-surface realizations $\{\widehat{C}_n^j(s_i)\}_{j=1}^N$ at each fixed $s_i \in \mathcal{S}$, we regress across the volatility states at time t_n to complete the continuation surface. Specifically, for each s_i we compute

$$a_n^N(s_i) \in \arg \min_{a \in \mathbb{R}^{d_B}} \sum_{j=1}^N \left| \widehat{C}_n^j(s_i) - a \cdot \phi(v_{t_n}^j, (v'_{t_n})^j) \right|^2. \quad (19)$$

The resulting completed continuation-value surface on the grid is

$$\widehat{C}_n^N(s_i, v, v') := a_n^N(s_i) \cdot \phi(v, v'),$$

and the value approximation at time t_n is given by the Bermudan recursion

$$\widehat{V}_n^N(s_i, v, v') := \max\{h_n(s_i), \widehat{C}_n^N(s_i, v, v')\}.$$

Starting from $\widehat{V}_M^N(s_i, v, v') := h_M(s_i)$ and iterating backward for $n = M - 1, \dots, 1$ produces an approximation of the value and continuation surfaces on $\mathcal{S} \times D_v$. If we consider the Bermudan put option, then $h_n(s) = (K - s)^+$ for all exercise dates t_n .

7.4 Time-zero estimators (direct and low)

At time $t_0 = 0$ the volatility state (v_0, v'_0) is fixed, so the time-zero continuation value may be estimated by averaging the first-step pre-surface values:

$$\widehat{C}_0(s_i) := \frac{1}{N} \sum_{j=1}^N \widehat{C}_0^j(s_i), \quad i = 1, \dots, N_S,$$

and defining the *direct* estimator

$$\widehat{V}_{d,0}(s_i) := \max\{h_0(s_i), \widehat{C}_0(s_i)\}.$$

To obtain a (typically) low-biased estimator, we use the estimated continuation surfaces to define a suboptimal stopping rule on the exercise grid. For an off-grid state $(S_{t_n}, v_{t_n}, v'_{t_n})$, evaluate $\widehat{C}_n^N(S_{t_n}, v_{t_n}, v'_{t_n})$ by interpolating $\widehat{C}_n^N(\cdot)$ in s and forming the dot product with $\phi(v_{t_n}, v'_{t_n})$. Define the stopping index

$$\tau^N := \inf \left\{ n \in \{0, \dots, M\} : h_n(S_{t_n}) \geq \widehat{C}_n^N(S_{t_n}, v_{t_n}, v'_{t_n}) \right\},$$

with the convention that $\tau^N = M$ if the set is empty. Simulating N_{low} independent sample paths of the full state process on the exercise grid and applying this policy yields the *low* estimator

$$\widehat{V}_{l,0} := \frac{1}{N_{\text{low}}} \sum_{k=1}^{N_{\text{low}}} e^{-rt_{\tau_k^N}} h_{\tau_k^N}(S_{t_{\tau_k^N}}^{(k)}).$$

8 Conclusion

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